Step functions

The **Heaviside function**, also called the Heaviside step function, or unit step function, is a step function in the form

$$u_c(t) = \begin{cases} 0 & 0 \leq t < c \\ 1 & t \geq c \end{cases}$$

In addition to $u_c(t)$, we'll also sometimes use $u(t-c)$,

$$u(t-c) = \begin{cases} 0 & 0 \leq t < c \\ 1 & t \geq c \end{cases}$$

or $H(t)$ or $H(t-c)$ to represent the Heaviside function. In any case, it's the function with a jump discontinuity at $t = c$. We only consider $c \geq 0$, which we see from the condition for the first case.

There's some debate about the value of the Heaviside function at exactly $t = c$. Sometimes the function's value will be defined as 0 at $t = c$, sometimes as 1 at $t = c$, and sometimes even as $1/2$ at $t = c$. For now, which convention we use isn't going to make a huge difference, so we'll stick with a value of 1 at $t = c$, which is the way we defined the piecewise functions above.

Characteristics

Think about the unit step function as a switch that turns on at $t = c$. When $t < c$, or to the left of $t = c$, the function is "off" because it has a value of 0.

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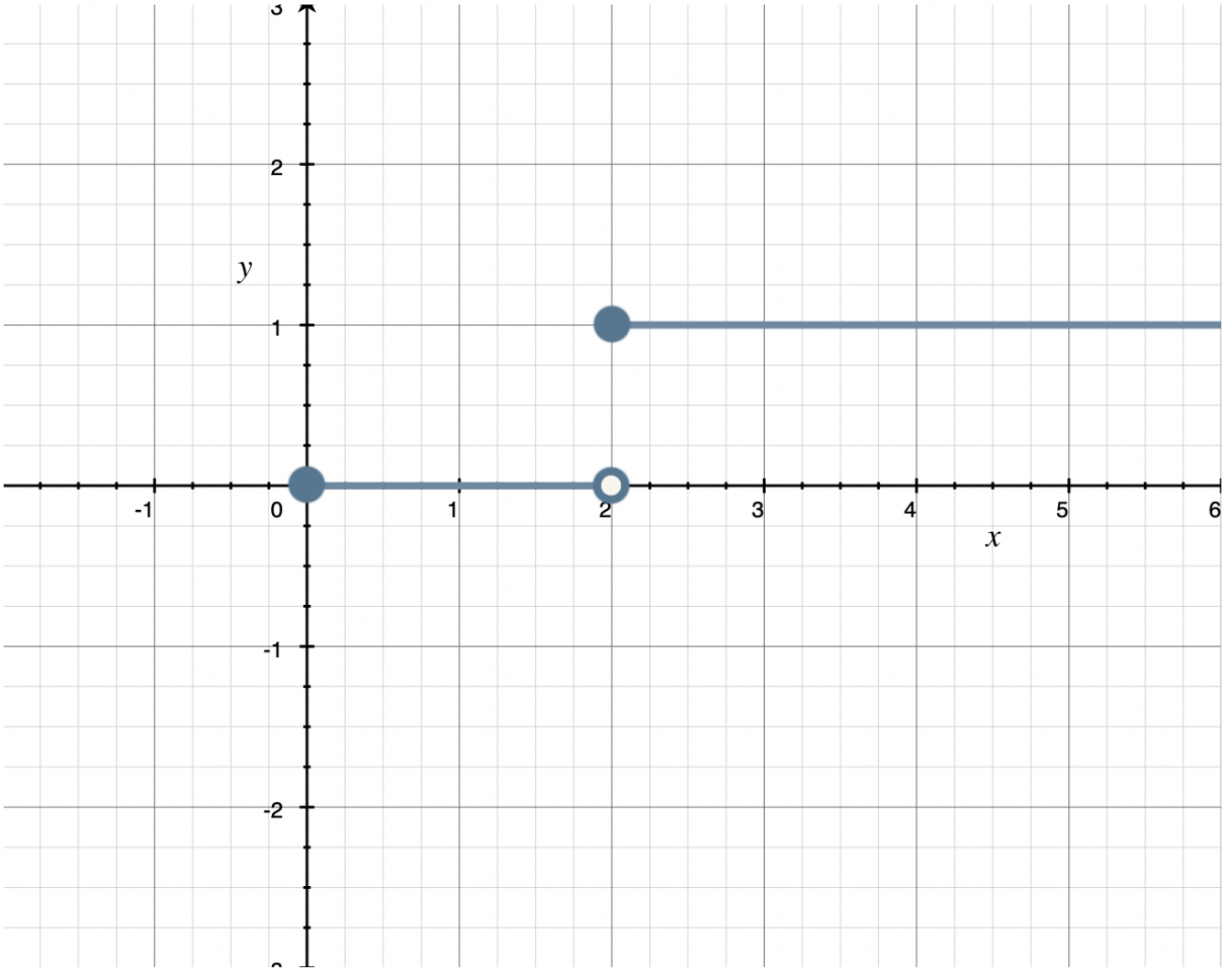
Then, as soon as we reach $t = c$, then function suddenly turns "on," and its value immediately jumps to 1.

In other words, we could say that the unit step function has these characteristics:

1. Its value is only ever 0 or 1; it never takes on any other value.
2. The change from 0 to 1 ("off" to "on") happens at $t = c$.
3. c could be any nonnegative value.

Below are the equation and the graph of the unit step function when $c = 2$.

$$u_2(t) = \begin{cases} 0 & 0 \le t < 2 \\ 1 & t \ge 2 \end{cases}$$



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Modifying the unit step function

While the unit step function only ever takes on values of 0 or 1, we can modify it to take other values. For instance, if we want a function that flips on to a value of 5, instead of a value of 1, we can simply multiply the unit step function by 5.

$$5u_c(t) = \begin{cases} 5(0) & 0 \leq t < c \\ 5(1) & t \geq c \end{cases}$$

$$5u_c(t) = \begin{cases} 0 & 0 \leq t < c \\ 5 & t \geq c \end{cases}$$

Notice how multiplying the left side of the equation by 5 required us to multiply both pieces on the right side by 5 as well. This new function $5u_c(t)$ will be "off" (have a value of 0) whenever $0 \leq t < c$, and then switch "on" and take a value of 5 as soon as $t = c$ (and will continue to take a value of 5 everywhere to the right of $t = c$).

We can also reverse the order of the switch, defining a function that starts "on" and then turns "off" at $t = c$, instead of the "off-to-on" pattern that we've been using up to this point.

Let's say we want the switch to be "on" at a value of 3, and then turn "off" to a value of 0 when we arrive at the switching point $t = c$. We'd write the equation of that function as

$$3 - 3u_c(t) = \begin{cases} 3 - 3(0) & 0 \leq t < c \\ 3 - 3(1) & t \geq c \end{cases}$$

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$$3 - 3u_c(t) = \begin{cases} 3 & 0 \le t < c \\ 0 & t \ge c \end{cases}$$

Here's a summary of these modifications:

Off-to-on

$$u_c(t) = \begin{cases} 0 & 0 \le t < c \\ 1 & t \ge c \end{cases}$$

Off-to-(on at n)

$$nu_c(t) = \begin{cases} 0 & 0 \le t < c \\ n & t \ge c \end{cases}$$

On-to-off

$$1 - u_c(t) = \begin{cases} 1 & 0 \le t < c \\ 0 & t \ge c \end{cases}$$

(On at n)-to-off

$$n(1 - u_c(t)) = \begin{cases} n & 0 \le t < c \\ 0 & t \ge c \end{cases}$$

Now we'd like to use these modifications to rewrite a piecewise function in terms of Heaviside functions, and vice versa. Let's work through an example.

Example

Express the piecewise function in terms of Heaviside functions.

$$f(t) = \begin{cases} -2 & 0 \le t < 1 \\ 4 & 1 \le t < 3 \\ 8 & 3 \le t < 5 \\ 12 & t \ge 5 \end{cases}$$

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Remember that a Heaviside function represents a switch, so we need one Heaviside function every time we make a switch in the value of $f(t)$. There are three switches, one from -2 to 4, one from 4 to 8, and one from 8 to 12, so we'll need three Heaviside functions in order to express $f(t)$.

We write a Heaviside function as $u_c(t)$, where the switch occurs at $t = c$. The switches in this function $f(t)$ are at $t = 1$, $t = 3$, and $t = 5$, which means the three Heaviside functions we'll need are $u_1(t)$, $u_3(t)$, and $u_5(t)$.

To express $f(t)$, we start with the idea that $f(t)$ has a value of -2 when it's "off." When the first switch at $t = 1$ turns on, the value of the function immediately shifts from -2 to 4 (up 6 units), so we can write

$$f(t) \approx -2 + 6u_1(t)$$

We use 6 as a coefficient on $u_1(t)$, since $-2 + 6 = 4$, and 4 is the value the function takes on after the first switch. Then moving from 4 to 8 is an increase of 4, which means we need to multiply $u_3(t)$ by 4,

$$f(t) \approx -2 + 6u_1(t) + 4u_3(t)$$

and moving from 8 to 12 is also an increase of 4, which means we need to multiply $u_5(t)$ by 4, and this will give us the full expression of $f(t)$ in terms of step functions.

$$f(t) = -2 + 6u_1(t) + 4u_3(t) + 4u_5(t)$$

If instead we'd been given this function for $f(t)$ in terms of unit step functions, we could still pull it apart and rewrite it as a piecewise function.

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The unit step functions show us that we have switches at $t = 1, 3, 5$, and the -2 gives us the "starting point." So immediately we can write

$$f(t) = \begin{cases} -2 & 0 \le t < 1 \\ & 1 \le t < 3 \\ & 3 \le t < 5 \\ & t \ge 5 \end{cases}$$

Then we consider the coefficients on the unit step functions, 6, 4, and 4. We'll add 6 to -2 to get $-2 + 6 = 4$ for the second piece,

$$f(t) = \begin{cases} -2 & 0 \le t < 1 \\ 4 & 1 \le t < 3 \\ & 3 \le t < 5 \\ & t \ge 5 \end{cases}$$

then add 4 again to get $4 + 4 = 8$ for the third piece,

$$f(t) = \begin{cases} -2 & 0 \le t < 1 \\ 4 & 1 \le t < 3 \\ 8 & 3 \le t < 5 \\ & t \ge 5 \end{cases}$$

and then add 4 again to get $8 + 4 = 12$ for the fourth piece.

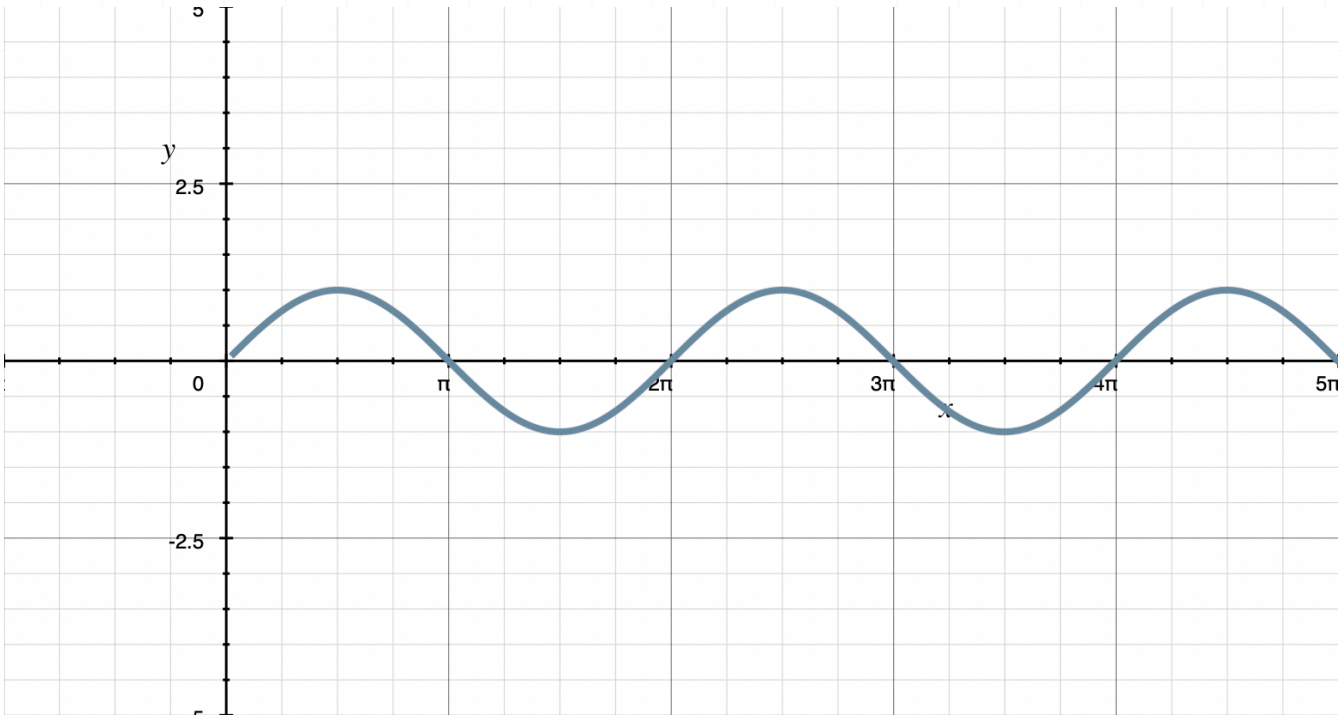
$$f(t) = \begin{cases} -2 & 0 \le t < 1 \\ 4 & 1 \le t < 3 \\ 8 & 3 \le t < 5 \\ 12 & t \ge 5 \end{cases}$$

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Non-constant "on" values

Up to now, we've been looking at step functions that take on some constant value when they're "on." But we also want to be able to turn on a function that takes on values that aren't constant.

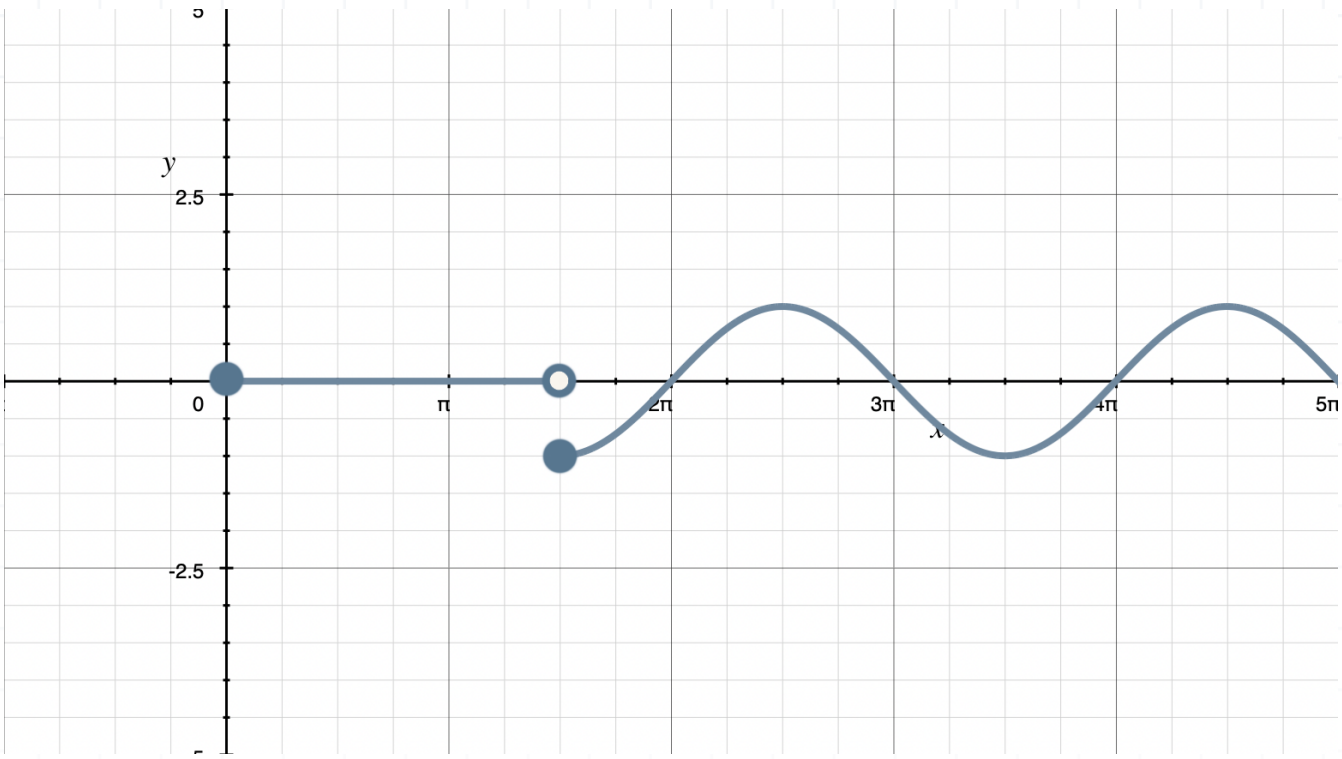
For example, let's consider the sine function $f(t) = \sin t$ just for $t \geq 0$.



We can use the unit step function to "turn off" a portion of this sine graph. Let's say we want to turn it off on $0 \leq t < 3\pi/2$. This "switch" is at $t = 3\pi/2$, which means we just need to multiply $\sin t$ by $u(t - 3\pi/2)$.

In other words, the graph of $f(t)u(t - c) = \sin t(u(t - 3\pi/2))$ is 0 (off) on $0 \leq t < 3\pi/2$, and then given by $\sin t$ (on) for $t \geq 3\pi/2$.

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To generalize this idea, we can say that the function $g(t)$,

$$g(t) = f(t)u(t-c)$$

$$g(t) = f(t)u_c(t)$$

is a function with a zero value (off) on the interval $0 \le t < c$, and then a value given by $f(t)$ (on) for $t \ge c$.

$$g(t) = \begin{cases} 0 & 0 \le t < c \\ f(t) & t \ge c \end{cases}$$

Let's do an example where we model this kind of function, where the value when the switch is "on" is given by some curve whose value varies.

Example

Express the piecewise function in terms of unit step functions.

$$g(t) = \begin{cases} t^2 + 4 & 0 \le t < 3 \\ 0 & t \ge 3 \end{cases}$$

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The first thing we notice about $g(t)$ is that it follows an "on-to-off" pattern instead of the default "off-to-on." Remember when we looked earlier at modifications of the step function, we said that this "(on at n)-to-off" pattern is given by

$$n(1-u_c(t)) = \begin{cases} n & 0 \le t < c \\ 0 & t \ge c \end{cases}$$

In other words, we can rewrite $g(t)$ as either of the following:

$$g(t) = (t^2+4)(1-u_3(t))$$

$$g(t) = (t^2+4)(1-u(t-3))$$

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